New Features Tour

Text styling

Important ideas stand out, *emphasis* adds tone,

and key terms mark vocabulary.

Short forms: **bold**, *italic*, underline.

Nesting works: **outer *inner* outer**.

Variables and calculations work inside styling:

answer is 6

Math works inside too: $\mathbf{\frac{1}{2}}$.

Numbers and sets

Real numbers: $\mathbb{R}$, naturals: $\mathbb{N}$, integers: $\mathbb{Z}$, rationals: $\mathbb{Q}$, complex: $\mathbb{C}$.

Long forms: $\mathbb{R}$ $\mathbb{N}$ $\mathbb{Z}$ $\mathbb{Q}$ $\mathbb{C}$.

The solution is $x \in \mathbb{R}$.

$$\begin{pmatrix} \mathbb{N} & \mathbb{Z} \\ \mathbb{Q} & \mathbb{C} \end{pmatrix}$$

Accents and derivatives

$\hat{x} + |y| + \tilde{z} + \dot{w}$

Segment AB: $\overline{AB}$

First derivative: $\frac{\mathrm{d}x^2}{\mathrm{d}x}$

Second derivative: $\frac{\mathrm{d}^2x^2}{\mathrm{d}x^2}$

Partial derivative: $\frac{\partial f}{\partial x}$

Determinants and sets

$$\det\left(\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\right)$$

$$\det\left(\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}\right)$$

Builder notation:

$$\{x \mid x > 0\}$$

Listing notation:

$$\{1, 2, 3\}$$

Two elements use commas:

$$\{a, b\}$$

Aligned equations

$$\begin{aligned} x + y &= 3 \\ x - y &= 1 \end{aligned}$$

$$\begin{aligned} x &= 1 \\ y &= 2 \end{aligned}$$

Calculations with functions

4 and 0 and 6.283185307179586.

The square root of 16 is 4.

Page break and image

After the break: a sine wave.

